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Keras3/Tensorflow environnement Hands-on 1 – Classification

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Preliminary remarks

These hands-on can be done on a cloud as CalculQuebec, Google Colab, Lab's Cloud, ... or on your own computer

For beginners : the recommended way is to start with CalculQuebec

For advanced users with CPU/GPU access

You can run on your specific hardware

Tips for installing the required environnement are [here](#).

<https://www.creatis.insa-lyon.fr/~grenier/?p=1716>

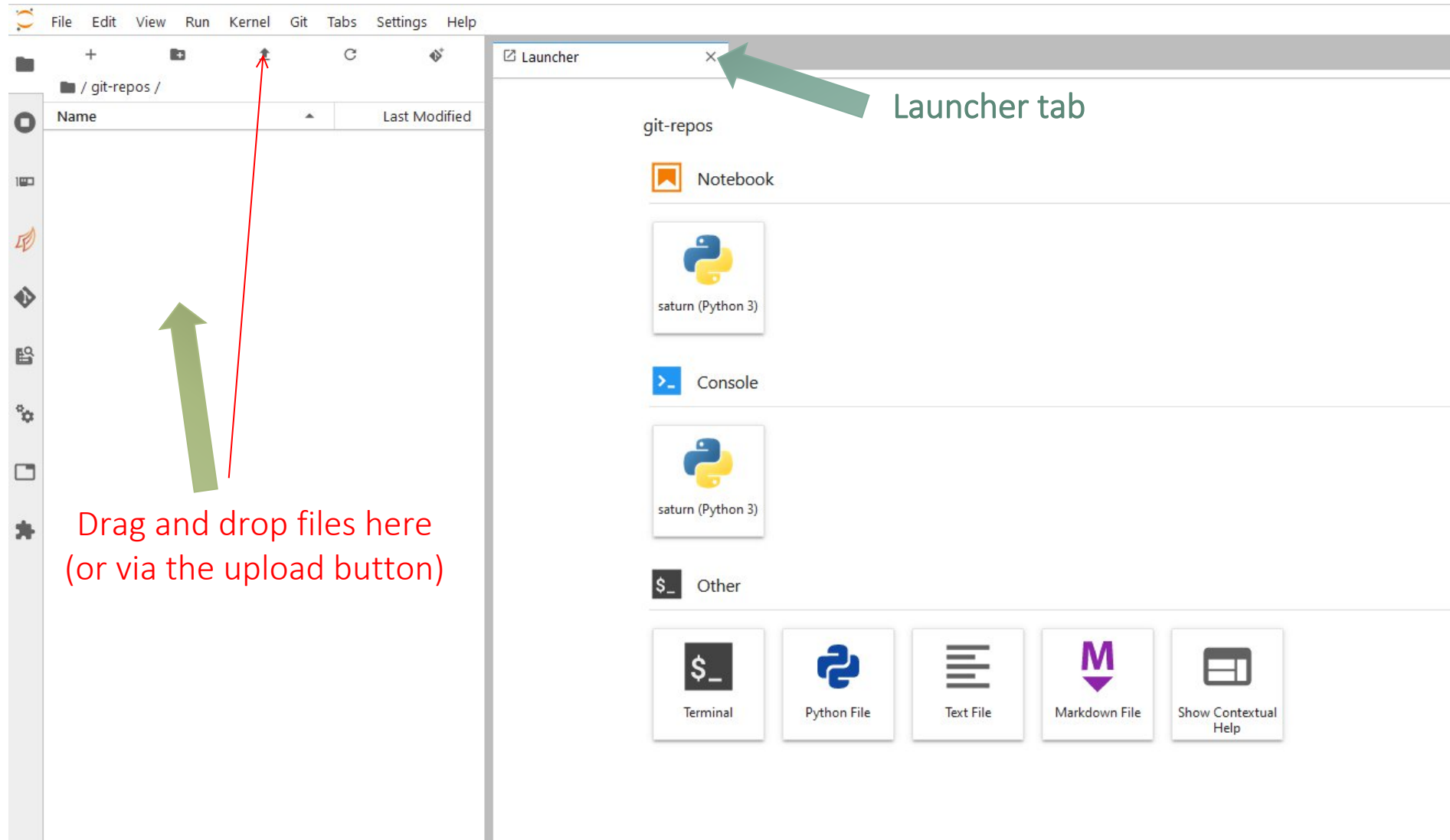
Steps required

1. Have an account on CalculQuebec Cloud and log-in
<https://jupyter.dmli.calculquebec.cloud/hub/login>
2. Create a compute node with
 1. Memory at 4096MB
 2. GPU 1x1G.5GB

Steps covered here

3. Start the jupyter-lab server and wait for it
4. Download the hands-on material into your jupyter-lab
5. Open a notebook
6. Cells philosophy and execution

3- Jupyter Lab interface (can change)



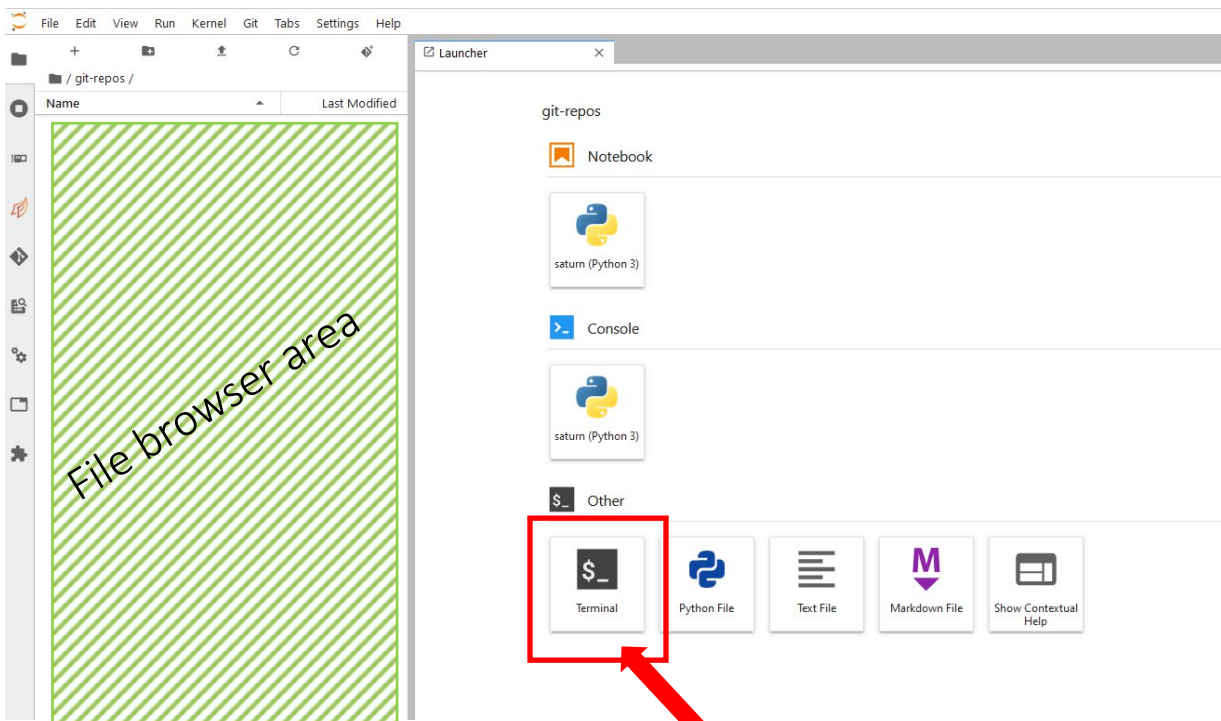
4- Download Hands-on materials : 2 ways

Way « Drag and drop » (recommended)

a- download on your computer the hands-on material as zip file (links? : next page!)

b- drag on drop it in the **file browser area**

c- unzip from jupyter lab (in a Terminal) —> next next page



Way « on server »

a- Open a **Terminal**

b- download from the terminal with :

```
wget
https://gitlab.in2p3.fr/thomas.grenier/tp
1ss_classification/-
/raw/master/TP_Classification_v06.zip
```

c- unzip from jupyter lab (in a Terminal)

—> next next page

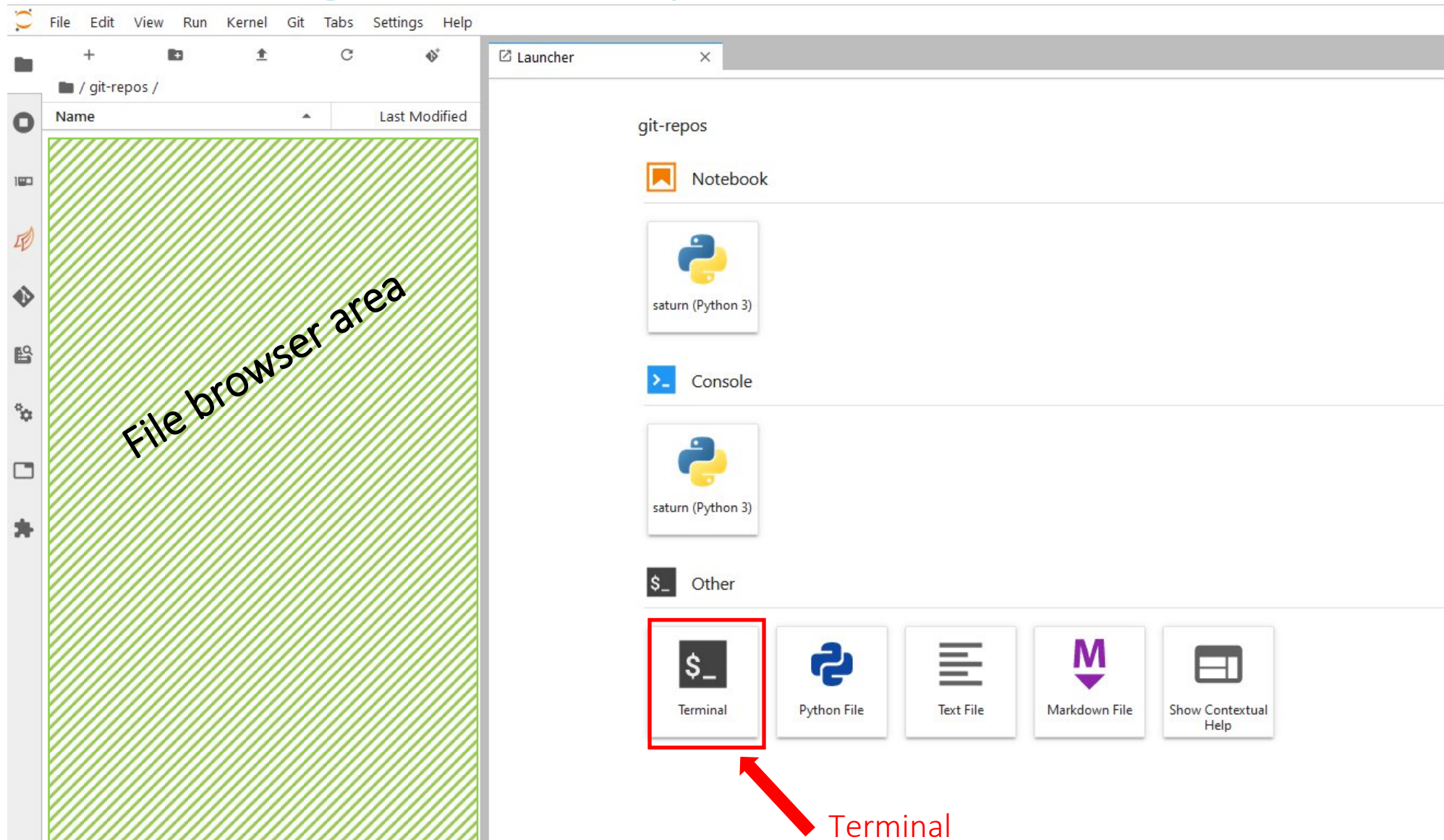
4a- Links to zip files

→ Hands-On 1 - Classification

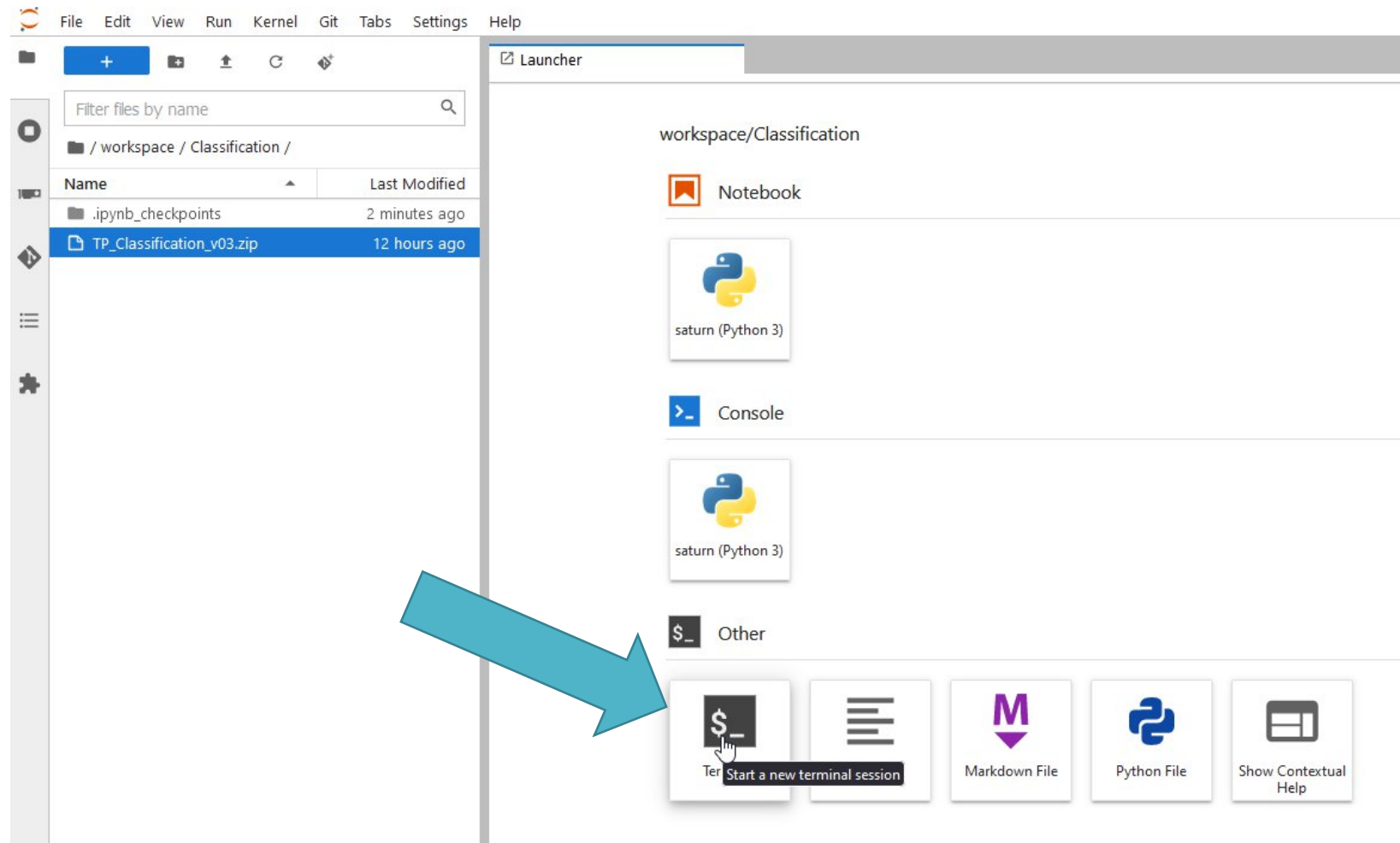
https://gitlab.in2p3.fr/thomas.grenier/tp1ss_classification/-/raw/master/TP_Classification_v06.zip

→ DON'T unzip this file on your computer
if you plan to use a Cloud as CalculQuebec!

4b- Drag and drop in the file browser area

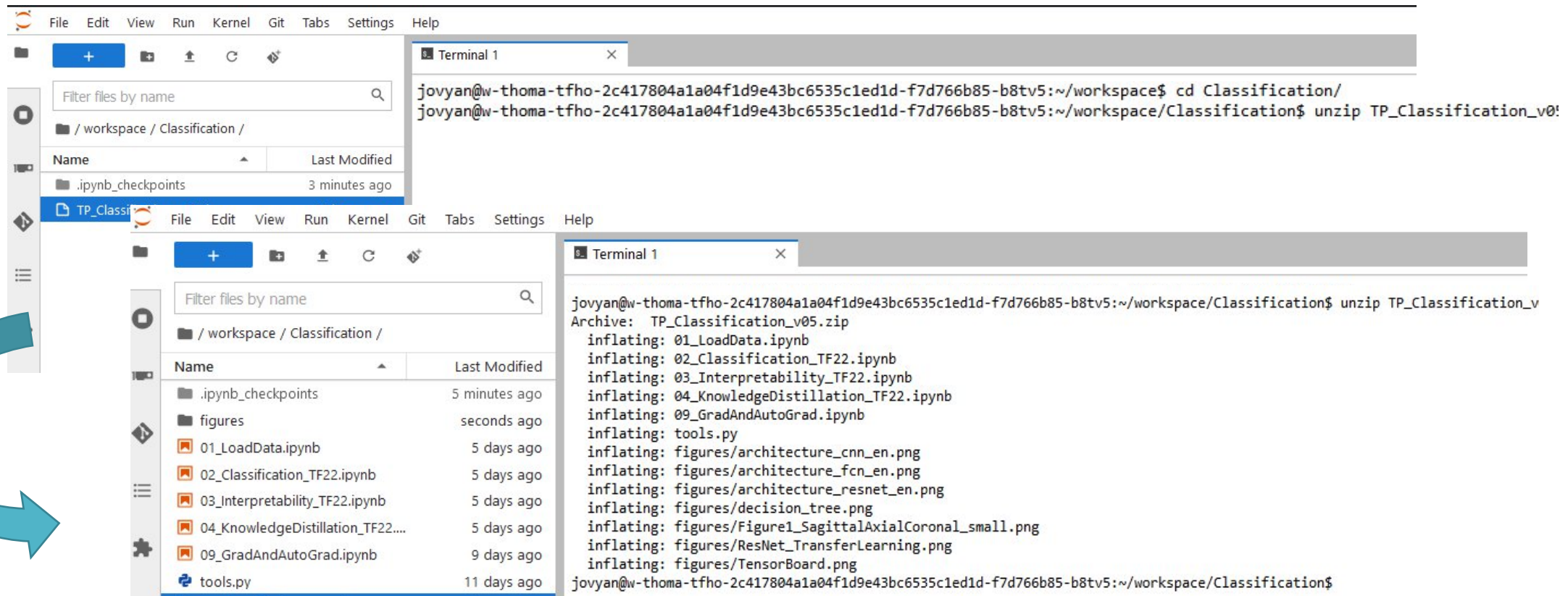


4c- unzip from jupyterlab open a Terminal



4c- unzip from jupyterlab unzip the file

Enter the command line: `unzip TP_Classification_v06.zip`
then « enter »



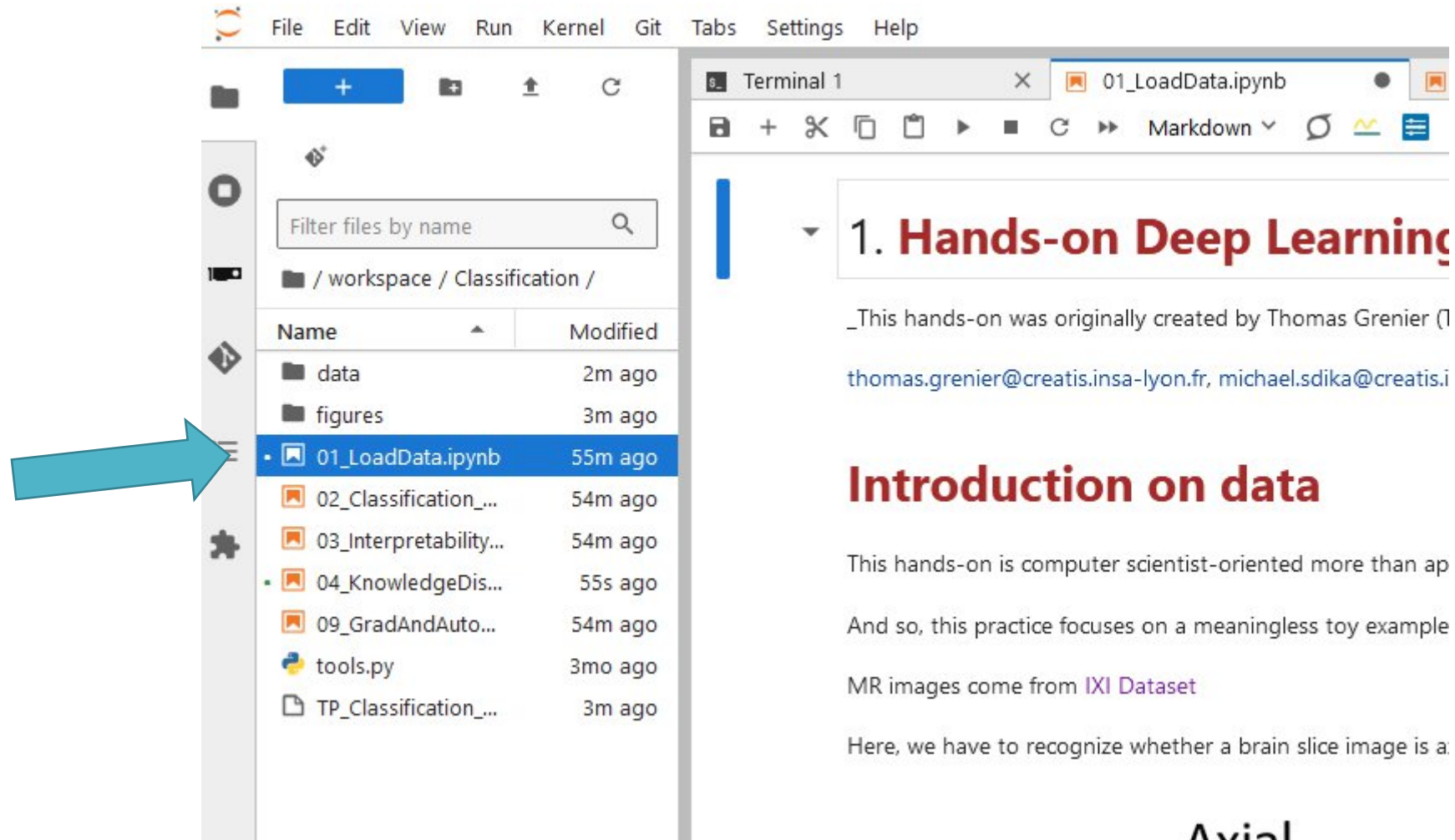
The top screenshot shows the JupyterLab interface with the file browser on the left and a terminal window on the right. The file browser shows the directory structure: `/ workspace / Classification /` with files `.ipynb_checkpoints` and `TP_Classification_v06.zip`. The terminal window shows the command `unzip TP_Classification_v06.zip` being executed.

The bottom screenshot shows the JupyterLab interface with the file browser on the left and a terminal window on the right. The file browser shows the directory structure: `/ workspace / Classification /` with files `.ipynb_checkpoints`, `figures`, `01_LoadData.ipynb`, `02_Classification_TF22.ipynb`, `03_Interpretability_TF22.ipynb`, `04_KnowledgeDistillation_TF22.ipynb`, `09_GradAndAutoGrad.ipynb`, and `tools.py`. The terminal window shows the output of the `unzip` command:

```
jovyan@w-thoma-tfho-2c417804a1a04f1d9e43bc6535c1ed1d-f7d766b85-b8tv5:~/workspace/Classification$ unzip TP_Classification_v05.zip
Archive: TP_Classification_v05.zip
  inflating: 01_LoadData.ipynb
  inflating: 02_Classification_TF22.ipynb
  inflating: 03_Interpretability_TF22.ipynb
  inflating: 04_KnowledgeDistillation_TF22.ipynb
  inflating: 09_GradAndAutoGrad.ipynb
  inflating: tools.py
  inflating: figures/architecture_cnn_en.png
  inflating: figures/architecture_fcn_en.png
  inflating: figures/architecture_resnet_en.png
  inflating: figures/decision_tree.png
  inflating: figures/Figure1_SagittalAxialCoronal_small.png
  inflating: figures/ResNet_TransferLearning.png
  inflating: figures/TensorBoard.png
jovyan@w-thoma-tfho-2c417804a1a04f1d9e43bc6535c1ed1d-f7d766b85-b8tv5:~/workspace/Classification$
```

5- Select and open the desired notebook

As example, double clic on : 01_LoadData.ipynb



The screenshot shows the JupyterLab interface. On the left, the file explorer displays the directory structure: / workspace / Classification /. The file list includes:

Name	Modified
data	2m ago
figures	3m ago
01_LoadData.ipynb	55m ago
02_Classification_...	54m ago
03_Interpretability...	54m ago
04_KnowledgeDis...	55s ago
09_GradAndAuto...	54m ago
tools.py	3mo ago
TP_Classification_...	3m ago

A blue arrow points to the file 01_LoadData.ipynb. The main editor area shows the notebook content, which includes a section titled "1. Hands-on Deep Learning" and an introduction to the data.

1. Hands-on Deep Learning

_This hands-on was originally created by Thomas Grenier (thomas.grenier@creatis.insa-lyon.fr, michael.sdika@creatis.i

Introduction on data

This hands-on is computer scientist-oriented more than ap

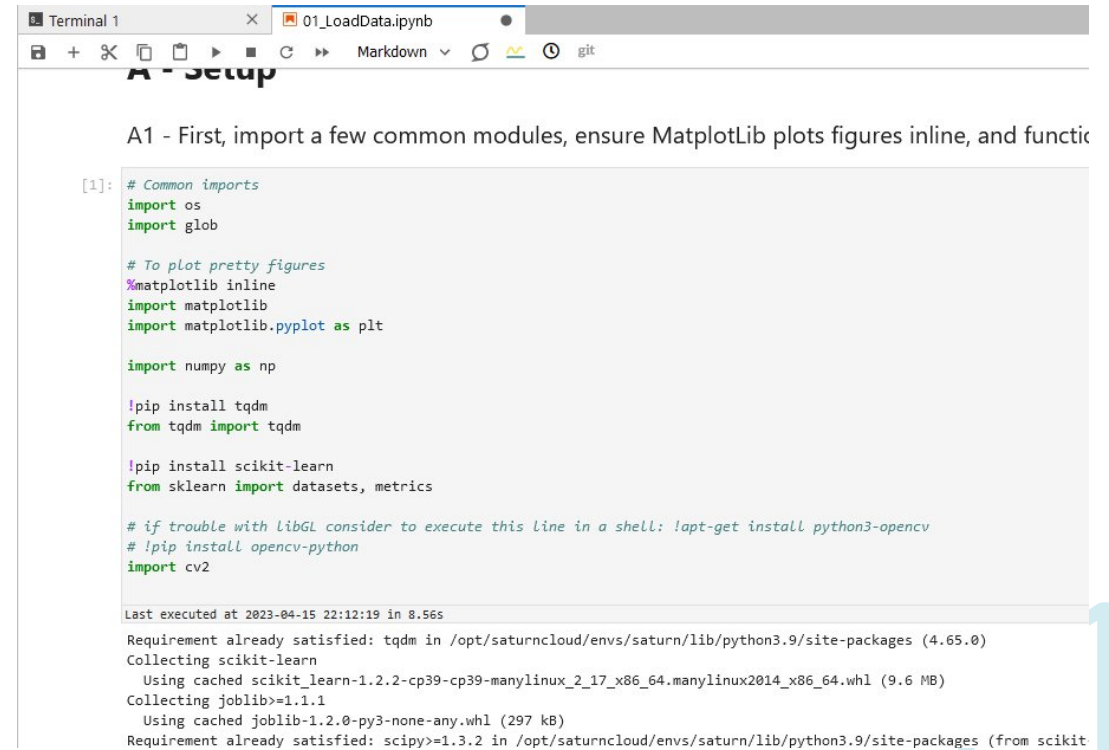
And so, this practice focuses on a meaningless toy example

MR images come from [IXI Dataset](#)

Here, we have to recognize whether a brain slice image is a.

6- Cells and running code

- Notebooks are made of cells : they can be text or python code
- Python code can be executed from the jupyter lab with play button 
- Outputs are added in a new cell after the code executed
- A cell can be executed many times
- Code of a cell can be changed
- Generally, a cell requires that the preceding cells have been executed before



Terminal 1 | 01_LoadData.ipynb | Markdown | git

A1 - First, import a few common modules, ensure Matplotlib plots figures inline, and function

```
[1]: # Common imports
import os
import glob

# To plot pretty figures
%matplotlib inline
import matplotlib
import matplotlib.pyplot as plt

import numpy as np

!pip install tqdm
from tqdm import tqdm

!pip install scikit-learn
from sklearn import datasets, metrics

# if trouble with LibGL consider to execute this line in a shell: !apt-get install python3-opencv
# !pip install opencv-python
import cv2
```

Last executed at 2023-04-15 22:12:19 in 8.56s

Requirement already satisfied: tqdm in /opt/saturncloud/envs/saturn/lib/python3.9/site-packages (4.65.0)
Collecting scikit-learn
Using cached scikit_learn-1.2.2-cp39-cp39-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (9.6 MB)
Collecting joblib=1.1.1
Using cached joblib-1.2.0-py3-none-any.whl (297 kB)
Requirement already satisfied: scipy>=1.3.2 in /opt/saturncloud/envs/saturn/lib/python3.9/site-packages (from scikit